Level-Aware Consistent Multilevel Map Translation From Satellite Imagery

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***Abstract*— With the rapid development of remote sensing technology, the quality of satellite imagery (SI) is getting higher, which contains rich cartographic information that can be translated into maps. However, existing methods either only focus on generating single-level map or do not fully consider the challenges of multilevel translation from satellite imageries, i.e., the large domain gap, level-dependent content differences, and main content consistency. In this article, we propose a novel level-aware fusion network for the SI-based multilevel map generation (MLMG) task. It aims to tackle these three challenges. To deal with the large domain gap, we propose to generate maps in a coarse-to-fine way. To well-handle the level-dependent content differences, we design a level classifier to explore different levels of the map. Besides, we use a map element extractor to extract the major geographic element features from satellite imageries, which is helpful to keep the main content consistency. Next, we design a multilevel fusion generator to generate a consistent multilevel map from the multilevel preliminary map, which further ensures the main content consistency. In addition, we collect a high-quality multilevel dataset for SI-based MLMG. Experimental results show that the proposed method can provide substantial improvements over the state-of-the-art alternatives in terms of both objective metric and visual quality.**

***Index Terms*— Large domain gap, level-dependent content differences, main content consistency, satellite imagery (SI).**

I. Introduction

THE multilevel map is widely used in the transportation industry, express delivery industry, personal travel, etc. Its generation relies on vector data, which is time-consuming to collect and update [1]. Meanwhile, with the rapid development of the satellite imagery (SI) technology, the quality of SI images has also become higher. We can use satellites, drones, and airplanes to collect SI images of the areas we need timely, which usually have high spatial resolution and contain rich cartographic information. These make it possible to translate SI images into maps quickly and conveniently via deep learning.

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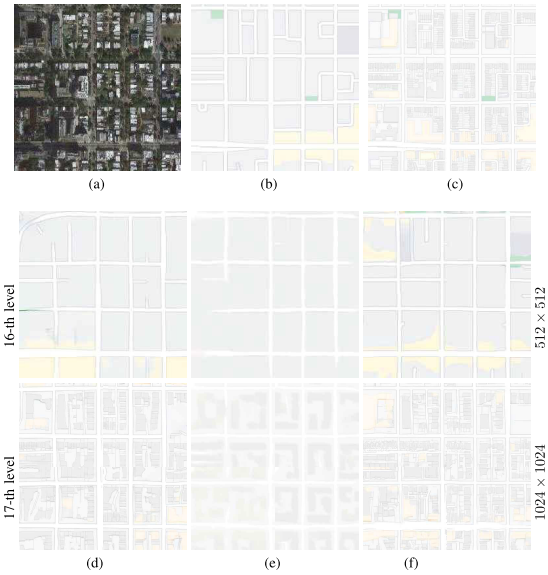

Fig. 1. SI-based MLMG is at different levels. (a) SI image. (b) and (c) Corresponding ground-truth (GT) map at the 16th and 17th levels. It can be seen that there are differences between maps at different levels, but there are also similarities, and pixel distribution varies greatly between SI images and maps. (d)–(f) Results from the existing SI-based single-level map generation methods (Pix2PixHD [2] and SMAPGAN [3]) and ours. It can be seen that our method provides high accuracy.

Recently, some works use deep learning to generate map image from SI image [3], [4], [5], [6], [7]. However, some existing methods [3], [4], [5] are limited to single-level map, which provide maps on a single scale and easily lead to the problem of content inconsistency when applied to multilevel map [see Fig. 1(d) and (e)]. Besides, other methods [6], [7] are designed for the SI-based multilevel map generation (MLMG). However, they do not adequately analyze the challenges of the SI-based MLMG, which limits their accuracy. To be specific, most existing methods fail to deal with the following three challenges.

1) Large domain gap: There is a large domain gap between SI image and map, e.g., pixel distribution is very different in SI image and map [see Fig. 1(a) and (b)].

2) Level-dependent content differences: Maps at different levels have different scales and their content [e.g., size of map elements (MEs)] is adjusted and optimized in terms of their scales, which makes the content at each level slightly different [see Fig. 1(b) and (c)].

3) Main content consistency: Maps at different levels provide information about the same geographic area,

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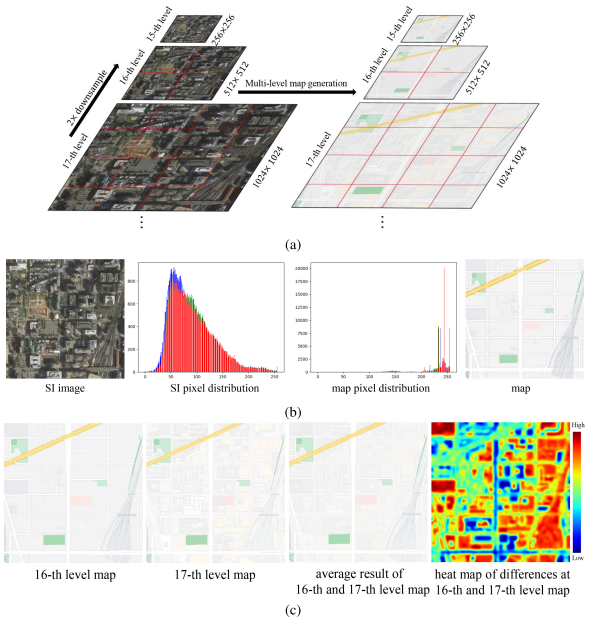

Fig. 2. Illustration of the SI-based MLMG, large domain gap, level-dependent content differences, and main content consistency. (a) Overview of the SI-based MLMG. The multilevel map is a set of maps with a pyramid structure. (b) Pixel distributions of the SI image and map show the large domain gap between the SI image and the map. (c) Average result looks like the 16th and 17th level maps. It means that there is consistency between the major geographic elements of the different levels, i.e., main content consistency. The heat map is obtained by mapping the absolute value of pixel difference between the 16th and 17th level maps. Blue color denotes less difference in value, i.e., main content consistency. Red color denotes greater difference in value, i.e., level-dependent content differences.

and thus, they should be consistent in the main content [see Fig. 1(b) and (c)], i.e., the main MEs in the same geographic area should be the same.

To generate a high-quality multilevel map from SI image, in this article, we propose a level-aware fusion network to tackle the above challenges. It consists of a level classifier, an ME extractor, a render (RD) generator, and a multilevel fusion (MLF) generator. As for the large domain gap, the huge difference in pixel distribution between SI image and map makes it difficult for the generator to generate a high-quality map, as shown in Fig. 2(b). Thus, we propose to generate maps in a coarse-to-fine way. Specifically, we design the dual-generator architecture for the network, including RD generator and MLF generator. We use the RD generator to generate preliminary maps, and then use the MLF generator to generate refined maps based on the preliminary maps. As for the level-dependent content differences, maps at different levels display different geographic elements. As shown in Fig. 2(c), the red part of the heat map shows the difference between maps at different levels. We design a level classifier to ensure the level-dependent content differences. It can discriminate maps from different levels and force the networks to generate a map with level-specific content. As for the main content consistency, we need to ensure that the major geographic elements of maps at different levels are displayed in the corresponding positions. As shown in Fig. 2(c), the blue part of the heat map shows the same geographic elements between different levels. We design an ME extractor and an MLF generator to tackle these problems. The ME extractor aims to extract features of major geographic elements (e.g., main roads, waters, and background) from SI image. These features can help the generator to pay more attention to these geographic elements, which can alleviate the lack of major geographic elements at all the levels. The MLF generator aims to fusion the $(k-1)$th and kth level preliminary maps to generate the kth level refined map. We perform the fusion operation at each level to further ensure the main content consistency.

In addition, we collect a large-scale SI-based MLMG dataset with equal amounts of data at four levels (15, 16, 17, and 18) to ensure that training at all the levels is balance. To keep the diversity of data, the dataset is collected from the United States (US) and China, two countries with large differences in SI image and map image. For each country, we provide 8000 samples for training and 1700 samples for evaluation. Each sample consists of an SI image, a corresponding map image, a level label, and an element label.

In summary, our main contributions are as follows.

1) We present a level-aware fusion network to generate a high-quality multilevel map from SI image only.
2) We design an RD generator, level classifier, ME extractor, and MLF generator to tackle challenges of large domain gap, level-dependent content differences, and main content consistency.
3) We collect an SI-based MLMG dataset, which can serve as a benchmark to evaluate and improve the MLMG method.

The remainder of this article is organized as follows. Section II reviews the related work. Section III introduces the motivation and the details of our methods. Section IV introduces our collected dataset (MLMG). In Section V, we provide the experimental results and comparisons. Finally, the conclusion is drawn in Section VI.

II. Related Work

In this section, we first briefly review the SI-based high-level visual tasks, then introduce the existing studies on the SI-based map generation in detail, which include the SI-based single-level map generation and MLMG, and finally present the existing public datasets for SI-based map generation.

A. SI-Based High-Level Visual Task

With the development of remote sensing technology, the acquisition of high-quality SI images is simpler and more convenient. This facilitates methods based on deep learning, which require high-quality data for training. The images of high-quality data are generally the map of Mercator projection [8]. Suppose a cylinder with the same direction as the axis of the Earth cuts or cuts the Earth, project the longitude and latitude network on the cylindrical surface according to the equiangular conditions and unfold the cylindrical surface into a plane to obtain the projection. After the planar projection is rasterized, the SI used for map generation is obtained. The SI-based high-level visual tasks include SI-based semantic segmentation, SI-based road extraction, SI-based map

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generation, etc. There are many categories and small sizes of geographical elements in SI images, which remains a challenge for SI semantic segmentation. The existing SI-based semantic segmentation methods [9], [10], [11], [12] mainly solve the problem of difficult semantic segmentation of geographical elements in SI images. The existing works [13], [14], [15], [16], [17], [18], [19] focus on extracting roads from SI images. The task of SI-based road extraction is to extract roads from SI images. They use the geometric and topological features of roads to assist road extraction. The existing SI-based target detection methods [20], [21], [22] mainly solve the problem of detecting small and dense targets with arbitrary rotation angles in SI images. Besides, there are also some SI-based map generation methods [3], [4], [5], [6], [7], which make it possible to generate maps quickly and conveniently.

B. SI-Based Map Generation

The SI-based map generation is an essential image-to-image translation task. It aims to convert the SI image into a readable map. However, the SI-based map generation requires additional consideration of the large domain gap between SI images and maps and solves this problem to improve the result. There are some classic methods [2], [23], [24], [25], [26] for the image-to-image translation task. However, they are not specifically designed for the SI-based map generation task and do not consider the large domain gap between SI images and maps. These make most of their results inferior to those methods that are specifically designed for the translation from SI images to maps. There are some methods [3], [4], [5] designed specifically for the SI-based map generation tasks. GeoGAN [4] attempts to translate SI images to map images using cGAN [27]. Semi-supervised generation of styled map tiles based on the generative adversarial network (SMAPGAN) [3] is a semisupervised model that uses incompletely paired SI image–map data for training and generates high-quality maps. Fu et al. [5] propose CreativeGAN to generate more accurate map images using semantic information. However, these methods only consider the SI-based single-level map generation.

Chen et al. [7] propose a simple training strategy to generate the multilevel map. At the highest level (the kth level, when k is the largest), it uses the SI image to generate the map. At other levels, this strategy uses multiple separated models to learn the mapping relationship between the kth map and the $(k-1)$th map. This is the reason why it can only train on the dataset that collects map pairs. Obviously, the results of this strategy are limited by the results of the highest level, and the error accumulates along with the level. In addition, the training process is cumbersome because it needs to train a special model for each level. Liu et al. [6] propose a single-model MLMG network that can generate multilevel maps using a single model. The innovation of this network is that a map-level classifier is added after the discriminator which is similar to StarGAN [25]. This method takes the ready-made road location shape information as one of the inputs. So we cannot use this method to quickly generate the multilevel map only from the SI image. In addition, the multilevel map discussed in this method is not a real multilevel map, i.e., a multilevel map of the same area. Its map dataset is a map of different levels from different regions. This limits its use in the SI-based MLMG.

TABLE I

COMPARISON WITH THE EXISTING MAP DATASETS. MLMG PROVIDES LARGER NUMBER OF SAMPLES AND IN MORE LEVELS

Dataset	SI	Map	Label	Same Area	Level Balanced Data	Levels
Isola [23]	√	√	×	-	-	1
AIDOMG [5]	√	√	√	-	-	1
SA dataset [6]	√	√	×	×	×	5
MLMG (ours)	√	√	√	√	√	4

In general, the existing methods either only focus on a single level or cannot satisfy the needs of the multilevel map in the same area. Thus, we adequately analyze the challenges in the SI-based MLMG and propose an end-to-end method to generate the multilevel map only from the SI image.

C. Public Dataset for the SI-Based MLMG

The development of the SI-based map generation datasets is still in the preliminary stage, and there is no mature dataset for the SI-based MLMG yet. Isola et al. [23] provide a small dataset of SI image–map pairs, but the dataset is relatively small and only contains single-level data. With the development of SI technology, high-quality SI images are easier to obtain and the quality of datasets is getting higher. Fu et al. [5] collect and produce a large dataset, aerial image dataset for online map generation (AIDOMG). The AIDOMG dataset is sampled from nine regions, providing high-quality SI images, map images, and the corresponding semantic labels for single level. Liu et al. [6] provide an SI-based multilevel map dataset with annotated data, but this dataset does not have multilevels of data for the same region. The details of these datasets are shown in Table I. To ensure the quality of results at all the levels, level-balanced data should be provided for training, and multilevel maps of the same area should be provided for evaluation. None of the existing datasets can satisfy the requirements of the SI-based MLMG dataset. Thus, we collect and produce a large SI-based multilevel map dataset from Google Maps named MLMG, which contains an equal number of samples at four levels for training, and multilevel data of the same area for evaluation. This dataset contains high-quality SI images and map images to support the training of our method.

III. METHOD

In this section, we first formulate the MLMG task, and then describe the proposed model in detail.

A. Formulation

With the development of SI technology, high-quality SI images with rich geographical information could be quickly and simply collected. This facilitates the map generation task. Specifically, the SI-based MLMG task is to extract the geographical information in the SI image pyramid and generate the multilevel maps, as shown in Fig. 2(a). The SI image pyramid is obtained by performing a series of $2\times$ downsampling on a single high-resolution SI image in terms of the scale of the map zoom level. Compared with SI images, multilevel

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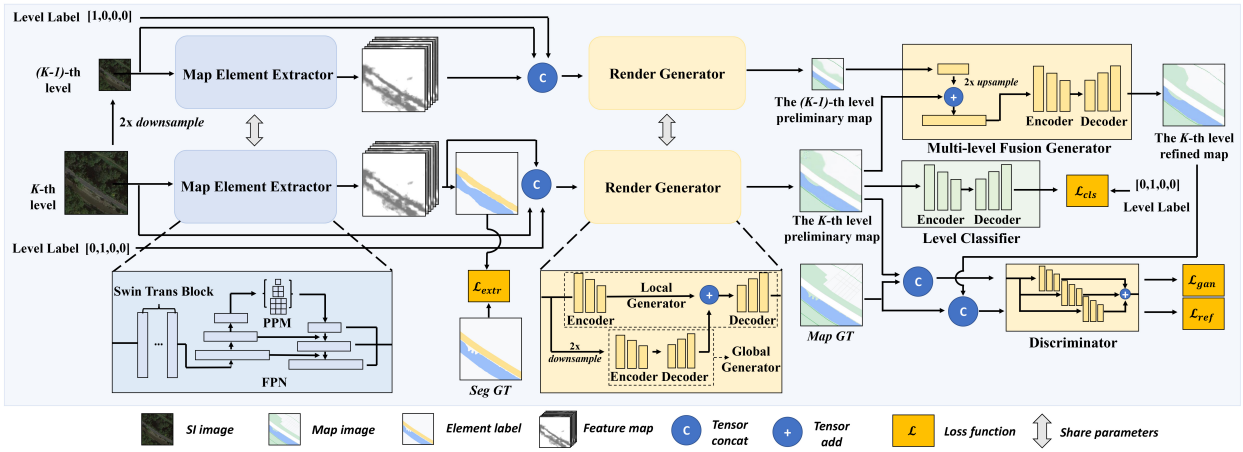

Fig. 3. Overview of our pipeline. The ME extractor is used to extract the major element feature. The function of the RD generator is to generate the preliminary map using the feature, level label, and SI image. Besides, the level classifier allows the RD generator to learn the differences at different levels. Finally, the MLF generator fuses preliminary maps of different levels to generate a more accurate refined map.

maps show a very different distribution of pixels compared with the SI image, and this is defined as a large domain gap [see Fig. 2(b)]. It is similar to the SI image pyramid, but it cannot be obtained by downsampling. Maps at different levels have some differences and similarities in the geographical elements, which are defined as level-dependent content differences and main content consistency [see Fig. 2(c)]. These characteristics make the task challenging.

To deal with the above three challenges in the task of SI-based MLMG, i.e., large domain gap, level-dependent content differences, and main content consistency, we propose a level-aware fusion network to generate multilevel maps from SI images that are considered to have large domain gap, level-dependent content differents, and main content consistency. It consists of level classifier, ME extractor, RD generator, and MLF generator. The overview of our proposed method is illustrated in Fig. 3.

B. Pipeline

Our pipeline is designed to tackle the large domain gap, level-dependent content differences, and main content consistency, generating accurate multilevel maps from SI images. As shown in Fig. 3, the ME extractor first extracts the features of major geographic element from the SI image, and then the features are sent to the RD generator. The 1-D one-hot vector of level label is repeated for each spatial location and is converted into 3-D one-hot format. Then, it is concatenated with the SI image and major geographic element features. The RD generator takes them as inputs to generate a preliminary map, which is coarse and needs to be refined. Then, the level classifier tries to classify the preliminary maps into their corresponding levels during training. This can make the results of the RD generator satisfy the level-dependent content differences. Besides, we use the RD generator and ME extractor to generate the $(k-1)$th level preliminary map. Finally, the MLF generator takes the $(k-1)$th and kth level preliminary maps as inputs to generate the kth level refined map. The refined map is the final output of our method.

TABLE II

DETAIL CONFIGURATION OF LEVEL CLASSIFIER. CONV: CONVOLUTIONAL LAYER, K: KERNEL SIZE, S: STRIDE SIZE, P: PADDING SIZE, O: OUTPUT CHANNEL

Layer	Layer Information	Norm	Activation
Conv	Conv(K4, S2, P1) O64	-	LeakyReLU
Conv	Conv(K4, S2, P1) O128	-	LeakyReLU
Conv	Conv(K4, S2, P1) O256	-	LeakyReLU
Conv	Conv(K4, S2, P1) O512	-	LeakyReLU
Conv	Conv(K4, S2, P1) O1024	-	LeakyReLU
Conv	Conv(K4, S2, P1) O2048	-	LeakyReLU
Conv	Conv(K4, S2, P1) O4096	-	LeakyReLU
Conv	Conv(K4, S2, P1) O4	-	-

The level classifier, ME extractor, RD generator, and MLF generator are trained jointly. Therefore, our model only needs to be trained once to generate multilevel maps. More details are provided in the following.

1) Level Classifier: Multilevel maps have level-dependent content differences among different levels. Thus, we propose a level classifier to constrain the RD generator to generate a better multilevel map. The level classifier classifies the preliminary map generated by the RD generator and forces the RD generator to learn the level-dependent content differences. It takes the preliminary map as input and outputs the predicted level label. The level classifier consists of nine layers of convolution with the leaky rectified linear unit (LeakyReLU) activation function [28], and the number of channels output by the last layer is the number of classes.

The level classifier can provide level constraints and guide the generator to generate maps that satisfy the characteristics of their respective levels. The level classifier can well-guarantee the level-dependent content differences for multilevel map. The details of the level classifier are shown in Table II.

2) ME Extractor: Maps at all the levels with major geographic elements are an important part of main content consistency, e.g., main roads, waters, and backgrounds. However, it is difficult for the generators to guarantee that these elements are shown at all the levels. To make the generated

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TABLE III

DETAIL CONFIGURATION OF ME EXTRACTOR. CONV: CONVOLUTIONAL LAYER, K: KERNEL SIZE, S: STRIDE SIZE, P: PADDING SIZE, O: OUTPUT CHANNEL, LN: LINEAR, WA: WINDOW ATTENTION, MLP: MULTILAYER PERCEPTRON, BN: BATCH NORMALIZATION

Layer	Layer Information	Norm	Activation
Conv	Conv(K4, S4, P0) O128	LN	-
TransformerBlock × 2	LN,WA,LN,MLP O128	-	-
TransformerBlock × 2	LN,WA,LN,MLP O256	-	-
TransformerBlock × 18	LN,WA,LN,MLP O512	-	-
TransformerBlock × 2	LN,WA,LN,MLP O1024	-	-
Conv × 3	Conv(K1, S1, P0) O512	BN	ReLU
Conv	Conv(K3, S1, P1) O512	BN	ReLU
Conv × 3	Conv(K1, S1, P0) O512	BN	ReLU
Conv × 3	Conv(K3, S1, P1) O512	BN	ReLU
Conv	Conv(K1, S1, P0) O3	-	-

TABLE IV

DETAIL CONFIGURATION OF GLOBAL GENERATOR. CONV: CONVOLUTIONAL LAYER, DECONV: DECONVOLUTIONAL LAYER, K: KERNEL SIZE, S: STRIDE SIZE, P: PADDING SIZE, O: OUTPUT CHANNEL, IN: INSTANCE NORMALIZATION

Layer	Layer Information	Norm	Activation
Conv	Conv(K7, S1, P3) O128	IN	ReLU
Conv	Conv(K3, S2, P1) O256	IN	ReLU
Conv	Conv(K3, S2, P1) O512	IN	ReLU
Conv	Conv(K3, S2, P1) O1024	IN	ReLU
ResBlk × 9	Conv(K3, S1, P1) O1024	IN	ReLU
	Conv(K3×3, S1, P1) O1024	IN	-
DeConv	DeConv(K3, S2, P1) O512	IN	ReLU
DeConv	DeConv(K3, S2, P1) O256	IN	ReLU
DeConv	DeConv(K3, S2, P1) O128	IN	ReLU

TABLE V

DETAIL CONFIGURATION OF LOCAL GENERATOR. CONV: CONVOLUTIONAL LAYER, DECONV: DECONVOLUTIONAL LAYER, K: KERNEL SIZE, S: STRIDE SIZE, P: PADDING SIZE, O: OUTPUT CHANNEL, IN: INSTANCE NORMALIZATION

Layer	Layer Information	Norm	Activation
Conv	Conv(K7, S1, P3) O64	IN	ReLU
Conv	Conv(K3, S2, P1) O128	IN	ReLU
ResBlk × 3	Conv(K3, S1, P1) O128	IN	ReLU
	Conv(K3×3, S1, P1) O128	IN	-
DeConv	DeConv(K3, S2, P1) O64	IN	ReLU
Conv	Conv(K7, S1, P3) O3	-	Tanh

images satisfy the main content consistency, we design an ME extractor. The function of the extractor is to extract the features of major geographic element. Then, the extracted features are sent into the generator to guide the generation of an accurate maps. The ME extractor takes the SI image as input and outputs element label which contains the major geographic elements. For the input SI images of different sizes, the map feature extractor will keep their sizes, and then output the features of the corresponding sizes. To better extract the feature, we adopt the latest Swin-Transformer [29] as the backbone. Then, we perform segmentation on the feature extractor by the backbone. Specifically, we place the pyramid pooling module (PPM) [30] after the bottom-up branch of the feature pyramid (FPN) [31] and before the top-down branch to improve the ability and obtain global information [30], [32]. Finally, a convolutional layer as the segmentation head is used to fuse the feature and generate the final element label. In addition, to get the accurate feature of major geographic elements, we add an auxiliary head fully convolutional network (FCN) [33] supervised by the element label follow pyramid scene parsing network (PSPNet) [30].

The features extracted by the ME extractor can provide information of the major geographic elements. It helps keep the main content consistency. The details of ME extractor are shown in Table III.

3) RD Generator: The RD generator is the fundamental part of our method. The RD generator is based on cGAN [27]. The function of the RD generator is to generate the preliminary map. The input of the generator includes the SI image, level label, and feature of major geographic elements. The output of the generator is the preliminary map. The generator has two subnetworks, i.e., a local generator and a global generator, which focuses on the whole image and the details of the image, separately. The global generator takes the two downsampled SI images, the level label, and the feature of major geographic elements as input to generate the low-resolution preliminary map. The local generator takes the original resolution SI image, level label, and feature of major geographic elements as input and fuses the result from the global generator.

The RD generator is one of the components of the dual-generator architecture, which is used to tackle the large domain gap. Using the RD generator, we can get the preliminary map, which can be used to generate the refined map. The details of RD generator are shown in Tables IV and V.

4) MLF Generator: The RD generator does not consider the relationship between different levels, and therefore, it is easy to cause inconsistencies in the multilevel map. In addition, to address the large domain gap, we propose to generate maps in a coarse-to-fine way. Thus, we propose an MLF generator to fuse preliminary maps of different levels to further generate a high-quality refined map. The MLF generator takes the kth and $(k-1)$th level preliminary maps as inputs and outputs the kth level refined map. The generator can be divided into two parts, the MLF head and the lightweight generator. The function of the MLF head is to fuse the multilevel preliminary maps to obtain the fusion features. To get effective fusion features, we first extract the features from the map using a convolutional layer, then interpolate the feature of the kth map, and finally, and add the features of the kth and $(k-1)$th map to obtain fusion features which fuse multilevel information. The lightweight generator is used to generate the refined map from fusion features and its structure is similar to RD generator. The MLF generator and RD generator share a discriminator, and the discriminator structure is shown in Table VI.

The RD generator is used to tackle the large domain gap. The MLF generator can further keep the main content consistency.

C. Loss Function

In this section, we introduce the loss functions used by each component.

1) Level Classifier: The level classifier is used in the classification task, and the cross-entropy loss [38] is widely used

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TABLE VI
DETAIL CONFIGURATION OF DISCRIMINATOR IN MULTISCALE DISCRIMINATOR. CONV: CONVOLUTIONAL LAYER, K: KERNEL SIZE, S: STRIDE SIZE, P: PADDING SIZE, O: OUTPUT CHANNEL, IN: INSTANCE NORMALIZATION

Layer	Layer Information	Norm	Activation
Conv	Conv(K4, S2, P2) O64	-	LeakyReLU
Conv	Conv(K4, S2, P2) O128	IN	LeakyReLU
Conv	Conv(K4, S2, P2) O256	IN	LeakyReLU
Conv	Conv(K4, S1, P2) O512	IN	LeakyReLU
Conv	Conv(K4, S1, P2) O1	-	-

in the fields of classification and semantic segmentation [25], [30], [39]. Thus, we use the cross-entropy loss as the loss function of the level classifier, in line with the functionality of the level classifier. The cross-entropy loss can be expressed as

$$\mathcal{L}_{\text{cls}} = \frac{1}{N}\sum_{k}\sum_{c=1}^{M} y_{\text{kc}}\log(p_{\text{kc}}) \tag{1}$$

where N is the number of samples, M is the number of levels, y is the real distribution, and p is the predicted distribution.

2) ME Extractor: The ME extractor essentially performs a pixel-wise semantic segmentation task. Thus, we use the cross-entropy loss [38] as the loss function of the ME extractor. The function of the ME extractor can be expressed as

$$\mathcal{L}_{\text{extr}} = \lambda_{\text{dec}}\mathcal{L}_{\text{dec}} + \lambda_{\text{aux}}\mathcal{L}_{\text{aux}} \tag{2}$$

where $\mathcal{L}_{\text{dec}}$ penalizes the distance between the final output and the actual label. $\mathcal{L}_{\text{aux}}$ penalizes the distance between the output of the FCN head [33] and the actual label. We set $\lambda_{\text{dec}} = 1$ and $\lambda_{\text{aux}} = 0.4$ to make the results more accurate.

3) RD Generator: For better SI-based map generation, we adopt content loss $\mathcal{L}_{\text{con}}$, adversarial loss $\mathcal{L}_{\text{adv}}$, perceptual loss $\mathcal{L}_{\text{per}}$, and feature-matching loss $\mathcal{L}_{\text{fm}}$. In general, the formula of the loss function of the RD generator can be expressed as

$$\mathcal{L}_{\text{gan}} = \lambda_{\text{con}}\mathcal{L}_{\text{con}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{per}}\mathcal{L}_{\text{per}} + \lambda_{\text{fm}}\mathcal{L}_{\text{fm}} \tag{3}$$

where we let $\lambda_{\text{adv}} = 1$ and $\lambda_{\text{con}} = \lambda_{\text{per}} = \lambda_{\text{fm}} = 10$ for loss balancing.

The content loss [23] $\mathcal{L}_{\text{con}}$ is used to calculate the pixel gap between the generated image and the real image, which facilitates the generation of more accurate images. We use mean absolute error as the content loss. The loss function is

$$\mathcal{L}_{\text{con}} = \frac{1}{N}\sum_{i=1}^{N}|G_{\text{RD}}(I_r) - I_m| \tag{4}$$

where N is the number of pixels of the image, I_m is the GT, I_r is the SI image, and G_{RD} is the RD generator.

The adversarial loss [40] $\mathcal{L}_{\text{adv}}$ makes the details of the image more realistic. The loss function is

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_{I_r, I_m}\left[\log D(I_r, I_m)\right] + \mathbb{E}_{I_r}\left[\log(1 - D(I_r, G_{\text{RD}}(I_r)))\right] \tag{5}$$

where I_r is the SI image, I_m is the GT, and D is the discriminator.

The perceptual loss [41] $\mathcal{L}_{\text{per}}$ can improve the overall visual quality of the generated map. The loss function is

$$\mathcal{L}_{\text{per}} = \sum_{i=1}^{n}\frac{1}{N_i}\left\|F^{(i)}(I_m) - F^{(i)}(G_{\text{RD}}(I_r))\right\|_1 \tag{6}$$

where F is the feature extraction network, i denotes the ith layer of the feature map from F, and n is the number of feature map layers. We use visual geometry group (VGG)16 as the feature extraction network.

The feature-matching loss [2] $\mathcal{L}_{\text{fm}}$ is useful to stabilize the training. It penalizes the difference between the real image and the generated image extracted from the discriminator. The loss function is

$$\mathcal{L}_{\text{fm}} = \sum_{i=1}^{n}\frac{1}{N_i}\left\|D^{(i)}(I_r, I_m) - D^{(i)}(I_r, G_{\text{RD}}(I_r))\right\|_1 \tag{7}$$

where n is the number of discriminator layers.

4) MLF Generator: The function of the MLF generator and the RD generator is to generate the map. Thus, we use the same loss function for the MLF generator. The loss function $\mathcal{L}_{\text{ref}}$ of the MLF generator is similar to the RD generator. It can be expressed as

$$\mathcal{L}_{\text{ref}} = \lambda_{\text{con}}\mathcal{L}_{\text{con}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}} + \lambda_{\text{per}}\mathcal{L}_{\text{per}} + \lambda_{\text{fm}}\mathcal{L}_{\text{fm}} \tag{8}$$

where we let $\lambda_{\text{adv}} = 1$ and $\lambda_{\text{con}} = \lambda_{\text{per}} = \lambda_{\text{fm}} = 10$ for loss balancing.

In general, the loss function of the entire model can be expressed as

$$\mathcal{L} = \lambda_{\text{gan}}\mathcal{L}_{\text{gan}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}} + \lambda_{\text{extr}}\mathcal{L}_{\text{extr}} + \lambda_{\text{ref}}\mathcal{L}_{\text{ref}} \tag{9}$$

where we set $\lambda_{\text{con}} = \lambda_{\text{cls}} = \lambda_{\text{extr}} = 1$ and $\lambda_{\text{ref}} = 4$ for loss balancing.

IV. OUR MLMG DATASET

The SI-based MLMG task requires paired multilevel data. However, due to the pyramid structure of paired multilevel data, the amount of data at each level decreases exponentially with the level number. It is the inherent level data imbalance problem in the multilevel map. The reason for the degradation of results at this level is the insufficient training data. Thus, the SI-based MLMG dataset should provide the level-balanced data for training. The level-balanced data are used to avoid the degradation of the result at the level where the number of samples is little. Besides, the SI-based MLMG dataset should also provide multilevel maps of the same area for evaluation. However, the existing public map datasets cannot satisfy these requirements of the SI-based MLMG dataset. Thus, we collect and produce a large SI-based MLMG dataset, called MLMG. The SI image and map of the MLMG dataset are collected from Google Maps, and we perform K-means [42] processing on the map to produce the element label. K-means clusters pixels with similar colors and finally gets the segmentation map of geographical elements. There are three classes per pixel for the element label: road, water, and background. A set of samples includes an SI image, a corresponding map image, and an element label.

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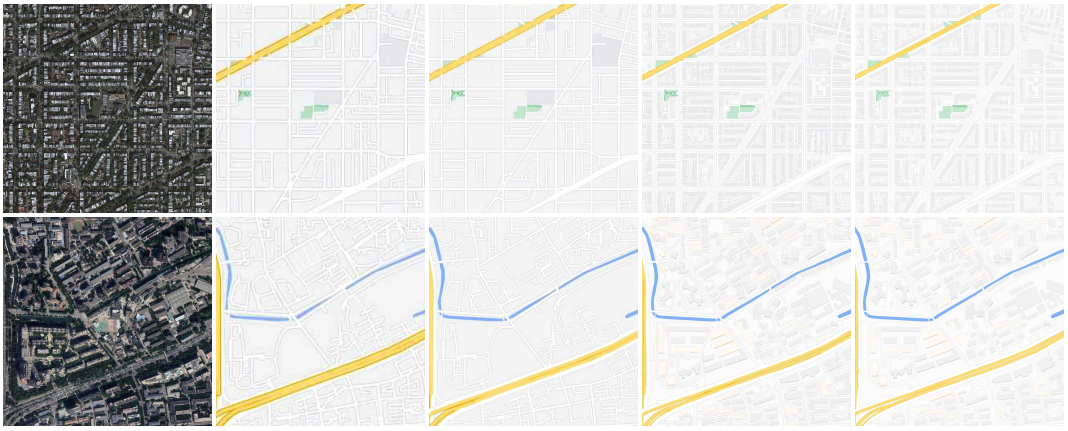

Fig. 4. Examples of our SI-based MLMG dataset. The first row shows the samples from the US, and the second row shows the samples from CN. (a) SI image. (b) Map at the 15th level. (c) Map at the 16th level. (d) Map at the 17th level. (e) Map at the 18th level.

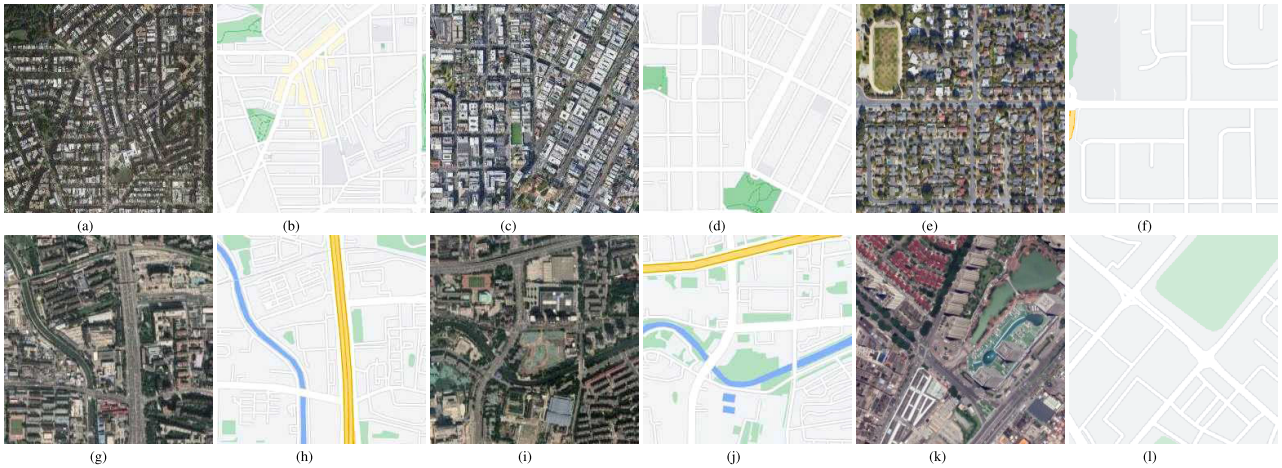

Fig. 5. Examples of different cities in different countries. We can see that the styles of SI images and map images of different cities in the same country are similar. (a) Washington SI. (b) Washington map. (c) Los Angeles SI. (d) Los Angeles map. (e) San Francisco SI. (f) San Francisco map. (g) Beijing SI. (h) Beijing map. (i) Tianjin SI. (j) Tianjin map. (k) Guangdong SI. (l) Guangdong map.

MLMG is, respectively, collected from the US and China. The SI image and map are very different in the US and China. As shown in Fig. 4, the roads in China tend to be wide and distributed in low density. There is no obvious parallel–perpendicular relationship between roads. The roads in the US tend to be straight and distribute in high density. The ground is neatly divided into neighborhoods to form a block. The SI-based MLMG dataset collected from the US cities is called MLMG-US. The SI-based MLMG dataset collected from the Chinese (CN) cities data is called MLMG-CN. MLMG-US and MLMG-CN have the same number of samples and contain the same levels. Both the datasets have SI images, map images, and element labels. The semantic categories of element labels in both the datasets are roads, water, and background.

We can collect multilevel maps directly from an area in Google Maps for evaluation, but they cannot be used directly for training because of data imbalance problem of the inherent level. The map number at the $(k-1)$th level is one-fourth of that at the kth level, one-sixteenth of that at the $(k+1)$th level, and so on. The unbalanced number of the map in the levels leads to the poor quality of the generated multilevel map, especially for the map at the $(k-1)$th. Fortunately, we find that the architectural styles and map styles of different major cities in the same country are similar, as shown in Fig. 5. We can see that buildings in Washington and Los Angeles are dense but orderly, and roads divide the ground into neat blocks. The roads in Beijing and Tianjin are relatively wide and the buildings are not dense. Thus, we can collect samples from major cities in the same country to supplement each level sample to the same number. For MLMG-US, we first collect a set of paired multilevel data from Washington at 15–18 levels. Then, we collect the corresponding high-quality SI images and map images from Los Angeles, San Francisco, New York, and Washington, for supplementing the 15–18 level data to equal. Similarly, for MLMG-CN, we first collect a set of multilevel data from Beijing with 15–18 level pairs. The rest are supplemented from Tianjin, Shanghai, Guangdong, and Beijing cities. The details of our MLMG dataset are shown in Table VII. Our dataset will be publicly available to the community.

V. Experiment

The proposed method is systematically evaluated on our MLMG dataset. In this section, we first provide our evaluation metrics and experimental details. Then, we compare

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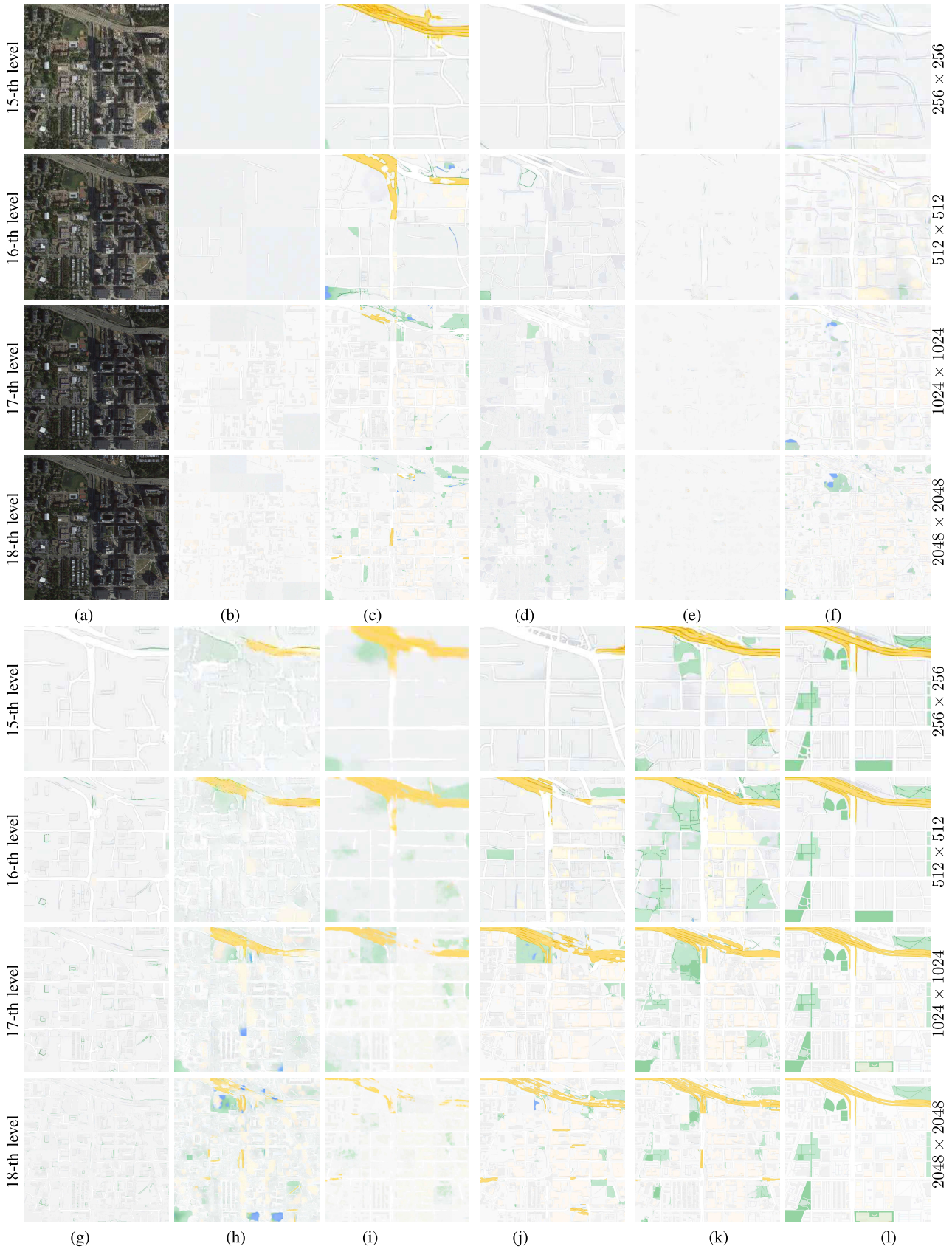

Fig. 6. Visualized results of MLMG-US generated by our method and counterparts. (a) SI image. (b) Pix2Pix [23]. (c) Pix2PixHD [2]. (d) CycleGAN [24]. (e) SPADE [34]. (f) SelectionGAN [35]. (g) TSIT [36]. (h) LPTN [37]. (i) SMAPGAN [3]. (j) CreativeGAN [5]. (k) Ours. (l) GT.

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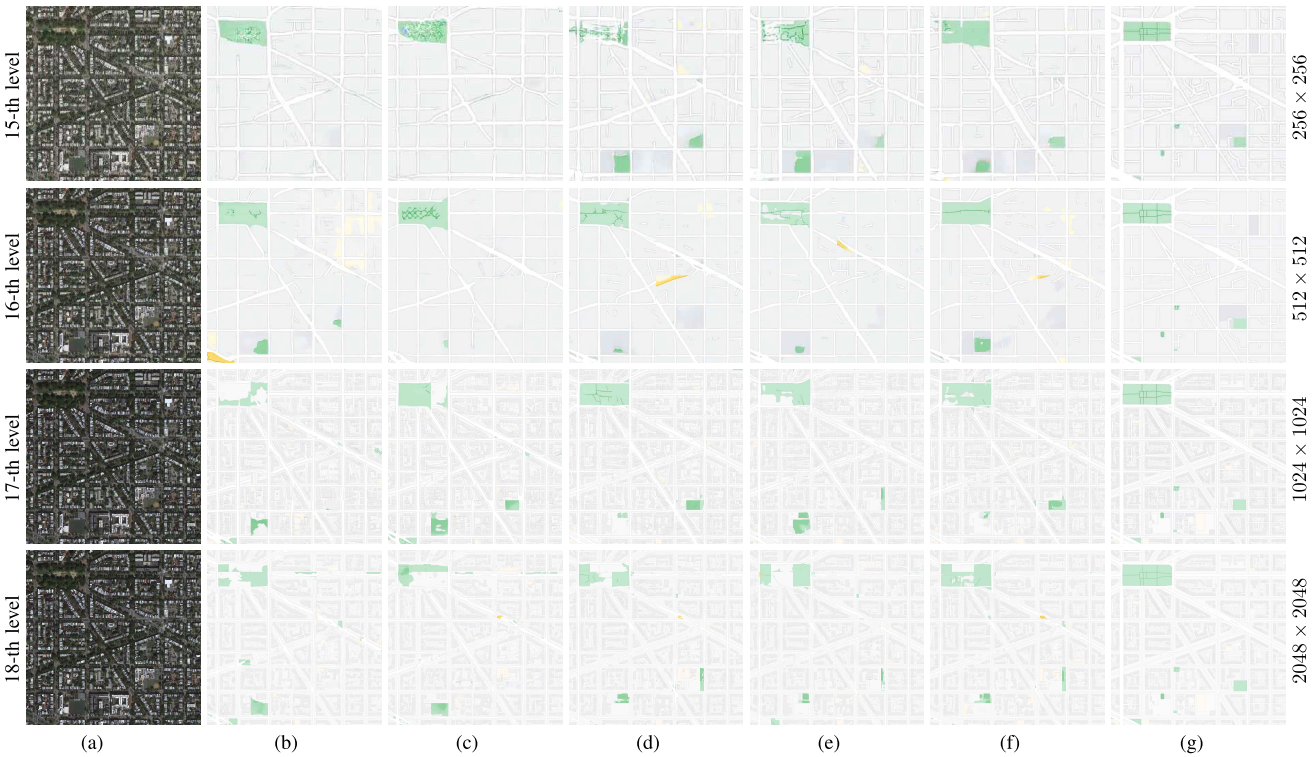

Fig. 7. Visualized results for ablation study on the MLMG-US dataset. Here are visualized results of our method after incrementally adding modules. (a) SI image. (b) Baseline. (c) Baseline. (d) Baseline. (e) Baseline. (f) Baseline. (g) GT + level classifier + level classifier + level classifier +l evel classifier + ME extractor + ME extractor + ME extractor + MLF generator + MLF generator w/o $(k-1)$th w/ $(k-1)$th.

and evaluate the existing methods. Subsequently, we perform ablation experiments to demonstrate the role of each component of the model and analyze the generated results. Finally, we analyze the results of the SI-based single-level map generation to demonstrate the generalizability of our method.

A. Evaluation Metrics

To quantitatively evaluate the generative effect of the model, we use Fréchet inception distance (FID) [43], Kernel maximum mean discrepancy (KMMD) [44], and Wasserstein distance (WD) [45] as metrics. These metrics are often used to evaluate generative adversarial network (GAN) generation results and work well in terms of discriminability, robustness, and efficiency [46].

In addition, we also use the classical evaluation metric, peak signal-to-noise ratio (PSNR) [47], as one of our metrics. Some researchers [48] have found that this metric does not match the visuals, but it still has a certain reference value for ablation experiments. The lower FID, KMMD, WD, and higher PSNR indicate better results.

B. Implementation Details

For our method, the network is trained for 100 epochs on a GeForce RTX 3090. We optimize the RD generator, the level classifier, and the MLF generator using the Adam [49] optimizer with exponential decay rates $(\beta_1, \beta_2) = (0.5, 0.999)$. The initial learning rate is 1×10^{-4} for the level classifier, and 2×10^{-4} for the RD generator and the MLF generator. We also use AdamW [50] with exponential decay rates $(\beta_1, \beta_2) = (0.9, 0.999)$ to optimize the ME extractor. The AdamW optimizer initializes the learning rate to 1.5×10^{-5} and uses the warm-up [51] technology. The warm-up step is set to 1500. We randomly initialize the RD generator, the level classifier, and the MLF generator. We initialize the Swin-transformer of the ME extractor with the pretrained weights on ImageNet [52] and randomly initialize the rest. For the competing methods, we adjust the batch size to the same size as our method to fit the memory of our computer and adjust the training iteration as the same as our method. As for other parameters, we use the default parameters in the source code without any modification.

TABLE VII
DETAILS OF OUR SI-BASED MLMG DATASET. IT CONSISTS OF SAMPLES FROM THE US REGION AND THE CN REGION

Zoom Level	Train Samples US | CN	Test Samples US | CN	Train + Test US | CN	Pixel Zoom
15	2,000 | 2,000	20 | 20	2,020 | 2,020	4.8 m/pixel
16	2,000 | 2,000	80 | 80	2,080 | 2,080	2.4 m/pixel
17	2,000 | 2,000	320 | 320	2,320 | 2,320	1.2 m/pixel
18	2,000 | 2,000	1,280 | 1,280	3,280 | 3,280	0.6 m/pixel
Total	8,000 | 8,000	1,700 | 1,700	9,700 | 9,700	**All 18,700**

C. Performance Comparison

To demonstrate the effectiveness of our method, we compare our method with the existing image-to-image translation methods (Pix2pix [23], Pix2pixHD [2], cycleGAN [24], SPADE [34], selectionGAN [35], two-stream image to image translation (TSIT) [36], and laplacian pyramid translation network (LPTN) [37]) and the SI-based map generation methods (SMAPGAN [3], Creative GAN [5]).

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TABLE VIII

QUANTITATIVE RESULTS OF DIFFERENT METHODS IN THE MLMG-US DATASET. "AVG." INDICATES THE MEAN VALUE OF THE RESULTS AT EACH LEVEL

Level	Method	FID↓	KMMD↓	WD↓	PSNR↑
15	Pix2Pix [23]	342.84	0.53236	16.267	20.726
	Pix2PixHD [2]	331.10	0.38806	14.358	20.908
	CycleGAN [24]	312.14	0.45603	14.692	20.725
	SPADE [34]	459.11	0.79792	20.047	20.468
	SelectionGAN [35]	337.83	0.58850	16.475	20.617
	TSIT [36]	284.17	0.43861	13.753	20.540
	LPTN [37]	351.61	0.4380	17.489	21.327
	SMAPGAN [3]	336.35	0.54771	16.338	**22.506**
	CreativeGAN [5]	267.37	0.29931	12.453	21.428
	Ours	**195.64**	**0.23759**	**11.014**	21.532
16	Pix2Pix [23]	256.58	0.42622	14.729	22.517
	Pix2PixHD [2]	206.16	0.28886	12.901	22.785
	CycleGAN [24]	237.79	0.36792	13.067	22.545
	SPADE [34]	351.68	0.66034	17.057	22.865
	SelectionGAN [35]	272.04	0.45795	14.653	22.702
	TSIT [36]	219.99	0.36471	13.048	22.543
	LPTN [37]	323.64	0.57608	16.683	23.454
	SMAPGAN [3]	292.50	0.51446	15.198	**24.819**
	CreativeGAN [5]	193.56	0.23249	11.674	22.947
	Ours	**154.18**	**0.19104**	**10.838**	23.488
17	Pix2Pix [23]	171.39	0.25765	12.025	23.878
	Pix2PixHD [2]	134.30	0.20902	11.092	23.682
	CycleGAN [24]	167.12	0.34208	12.496	23.076
	SPADE [34]	295.67	0.54238	14.779	24.000
	SelectionGAN [35]	239.45	0.39211	13.530	23.914
	TSIT [36]	137.50	0.30070	11.945	23.264
	LPTN [37]	253.00	0.45528	14.431	24.198
	SMAPGAN [3]	286.87	0.54847	15.712	**25.340**
	CreativeGAN [5]	129.51	0.17238	10.710	23.640
	Ours	**107.95**	**0.14169**	**9.813**	24.514
18	Pix2Pix [23]	138.45	0.28072	11.693	25.519
	Pix2PixHD [2]	110.35	0.25507	11.017	24.801
	CycleGAN [24]	138.92	0.36987	12.519	23.715
	SPADE [34]	224.64	0.51832	13.771	25.330
	SelectionGAN [35]	194.97	0.41975	13.236	25.442
	TSIT [36]	123.45	0.35038	12.035	24.167
	LPTN [37]	182.36	0.43861	13.672	24.599
	SMAPGAN [3]	246.88	0.55472	14.524	**27.005**
	CreativeGAN [5]	100.54	0.22727	10.719	24.799
	Ours	**78.44**	**0.17341**	**9.503**	25.970
Avg.	Pix2Pix [23]	227.31	0.37424	13.679	23.160
	Pix2PixHD [2]	195.48	0.28525	12.342	23.044
	CycleGAN [24]	213.99	0.38398	13.193	22.515
	SPADE [34]	332.78	0.62974	16.413	23.166
	SelectionGAN [35]	261.07	0.46457	14.474	23.169
	TSIT [36]	191.28	0.36360	12.695	22.628
	LPTN [37]	277.65	0.52357	15.569	23.394
	SMAPGAN [3]	290.65	0.54134	15.443	**24.918**
	CreativeGAN [5]	172.74	0.23286	11.389	23.204
	Ours	**134.05**	**0.18593**	**10.292**	23.876

TABLE IX

QUANTITATIVE RESULTS OF DIFFERENT METHODS IN THE MLMG-CN DATASET. "AVG." INDICATES THE MEAN VALUE OF THE RESULTS AT EACH LEVEL

Level	Method	FID↓	KMMD↓	WD↓	PSNR↑
15	Pix2Pix [23]	332.01	0.58042	16.077	24.540
	Pix2PixHD [2]	264.73	0.49512	14.251	23.948
	CycleGAN [24]	243.28	0.51064	14.741	23.718
	SPADE [34]	367.41	0.69403	17.401	23.663
	SelectionGAN [35]	323.33	0.56767	14.679	23.883
	TSIT [36]	254.89	0.51055	14.388	23.272
	LPTN [37]	320.06	0.58184	15.330	23.683
	SMAPGAN [3]	414.42	0.70441	17.143	**25.766**
	CreativeGAN [5]	225.38	0.40546	0.40546	24.403
	Ours	**189.01**	**0.34095**	**11.414**	24.385
16	Pix2Pix [23]	331.97	0.53578	14.521	27.426
	Pix2PixHD [2]	193.22	0.34912	12.300	25.669
	CycleGAN [24]	164.28	0.30605	11.972	25.114
	SPADE [34]	317.71	0.62258	15.110	26.604
	SelectionGAN [35]	272.06	0.44437	12.598	26.150
	TSIT [36]	193.88	0.39535	12.190	25.418
	LPTN [37]	292.19	0.57657	14.640	25.409
	SMAPGAN [3]	376.17	0.65042	15.743	**28.269**
	CreativeGAN [5]	157.25	0.26949	11.062	25.894
	Ours	**141.47**	**0.25077**	**10.805**	25.332
17	Pix2Pix [23]	210.13	0.34034	11.514	26.604
	Pix2PixHD [2]	169.32	0.35069	12.297	25.315
	CycleGAN [24]	114.20	0.27646	10.590	24.812
	SPADE [34]	305.92	0.65596	15.141	26.363
	SelectionGAN [35]	250.43	0.43441	12.480	25.581
	TSIT [36]	173.37	0.47257	12.962	25.183
	LPTN [37]	233.89	0.56163	14.378	24.921
	SMAPGAN [3]	331.03	0.68852	15.800	**27.545**
	CreativeGAN [5]	123.77	0.25430	10.641	25.472
	Ours	**105.27**	**0.19193**	**9.892**	25.361
18	Pix2Pix [23]	120.55	0.27553	10.093	27.054
	Pix2PixHD [2]	125.63	0.32556	11.593	26.255
	CycleGAN [24]	90.25	0.28707	10.442	25.756
	SPADE [34]	226.82	0.58484	13.959	27.163
	SelectionGAN [35]	194.92	0.45119	12.667	26.636
	TSIT [36]	134.08	0.43336	12.313	25.729
	LPTN [37]	181.25	0.53740	14.328	24.626
	SMAPGAN [3]	267.91	0.69492	15.331	**28.674**
	CreativeGAN [5]	90.98	0.24425	10.218	26.965
	Ours	**67.43**	**0.19328**	**9.549**	26.556
Avg.	Pix2Pix [23]	248.66	0.43302	13.051	26.406
	Pix2PixHD [2]	188.22	0.38012	12.610	25.297
	CycleGAN [24]	153.00	0.34506	11.936	24.850
	SPADE [34]	304.46	0.63935	15.403	25.948
	SelectionGAN [35]	260.18	0.47441	13.106	25.563
	TSIT [36]	189.05	0.45296	12.963	24.901
	LPTN [37]	256.85	0.56436	14.669	24.660
	SMAPGAN [3]	347.38	0.68457	16.004	**27.564**
	CreativeGAN [5]	149.34	0.29337	11.152	25.683
	Ours	**125.80**	**0.24423**	**10.415**	25.409

We train these models using MLMG-CN and MLMG-US, respectively, and separately. As shown in Tables VIII and IX, for GAN metrics such as FID, KMMD, and WD, our method achieves the best results at each level and the average results across levels. For PSNR, the results of our method are inferior to SMAPGAN. We need to note that the visual results of SMAPGAN in Fig. 6 are far inferior to our method. The generated results of SMAPGAN show blurred and unclear. Some researchers [48], [53] have found that a higher PSNR does not mean better visual results. The reason we keep this metric is that it has a certain reference value in the ablation experiment.

Fig. 6 shows the visualized results of our method and the comparison method. We can see that our method works very close to the GT at all the levels. The results of other methods have more breaks of road and noise at all the levels, especially at lower levels.

TABLE X

SIZE AND INFERENCE EFFICIENCY OF DIFFERENT MODELS

Method	Model Size	Inference Efficiency
Pix2pix	57 M	11.3 fps
Pix2pixHD	779 M	8.3 fps
CycleGAN	113 M	8.8 fps
SPADE	391 M	8.9 fps
SelectionGAN	233 M	84.0 fps
TSIT	1599 M	32.4 fps
LPTN	3.5 M	20.7 fps
SMAPGAN	113 M	6.2 fps
CreativeGAN	1008 M	5.2 fps
Ours	2004 M	4.0 fps

These results show that our method generates far better results than the compared methods in the SI-based MLMG and prove the effectiveness of our method. Though the parameters of the proposed method may be relatively large and its speed is not fast, as shown in Table X, its performance outperforms the existing methods by a large margin. It is worth noting

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TABLE XI
QUANTITATIVE RESULTS OF OUR METHOD ON THE MLMG-US DATASET FOR ABLATION STUDY

Level	Baseline	+Level Classifier	+ME Generator	+MLF Generator		FID↓	KMMD↓	WD↓	PSNR↑
				w/o $(k-1)$-th	w/ $(k-1)$-th				
Avg.	√	×	×	×	×	195.48	0.28525	12.342	23.044
	√	√	×	×	×	188.65	0.28144	12.250	23.104
	√	√	√	×	×	160.08	0.20580	10.815	23.654
	√	√	√	√	×	135.72	0.18697	10.414	23.747
	√	√	√	×	√	**134.05**	**0.18593**	**10.292**	**23.876**

TABLE XII
QUANTITATIVE RESULTS WITH DIFFERENT FORMAT OF LEVEL LABEL. THE RESULTS OF ONE-HOT HAVE HIGHER QUALITY THAN OTHER METHODS

Format	FID↓	KMMD↓	WD↓	PSNR↑
One-Hot	**134.05**	0.1859	**10.292**	**23.876**
Numerical label	135.69	**0.1839**	10.356	23.872
Soft Label	136.72	0.1915	10.408	23.713

TABLE XIII
QUANTITATIVE RESULTS WITH USING AUXILIARY FUNCTION AND NOT USING AUXILIARY FUNCTIONS. THE RESULTS OF USING AUXILIARY FUNCTION HAVE HIGHER QUALITY THAN NOT USING

Method	FID↓	KMMD↓	WD↓	PSNR↑
Ours w/ auxiliary loss function	**134.05**	**0.1859**	10.292	**23.876**
Ours w/o auxiliary loss function	136.34	0.1891	**10.264**	**23.876**

that the quality of the generated map is very important for the map generation task. Map generation with low quality but fast speed is meaningless.

D. Ablation Study

We verify the effectiveness of each component of our method. Table XI shows the results of the ablation experiments with the addition of the level classifier, the ME extractor, and the MLF generator.

1) Level Classifier: The role of the level classifier is to assist the generator to optimize the output mapping at each level. It is used to ensure the level-dependent content differences. As shown in Fig. 7(c), we can see that the visual results of the map generated after adding this classifier are better than that without this classifier. Specifically, the yellow background and roads that should not exist on the 16-level map are removed with the help of the level classifier. From Table XI, we can see that after adding the level classifier, all the metrics are improved compared with the baseline, especially the FID is increased by 6.83. By adding the level classifier, the results of each level are improved. The generated map at each level is more in line with the characteristics of its level.

2) ME Extractor: The role of the ME extractor is to extract the features of major geographic element, which is used to ensure the main content consistency. It allows the model to focus more on the generation of major geographic elements and generate a better quality map. As shown in Fig. 7(d), we can see that the continuity and integrity of major geographic elements are greatly improved after adding this extractor. Specifically, roads on the map are more coherent and vegetation locations are more accurate. The visual results get better when adding the ME extractor. From Table XI, we can see that all the results are greatly improved after adding the extractor. Specifically, FID is increased by 28.57, KMMD is increased by 0.07564, WD is increased by 1.435, and PSNR is increased by 0.55. By adding the level ME extractor, the

TABLE XIV
QUANTITATIVE RESULTS OBTAINED BY DIFFERENT METHODS. WE USE THE TRAINING SET OF MLMG-CN FOR TRAINING AND THE TEST SET OF MLMG-US FOR TESTING

Level	Method	FID↓	KMMD↓	WD↓	PSNR↑
15	Pix2Pix [23]	396.02	0.59515	17.601	21.119
	Pix2PixHD [2]	393.64	0.53683	16.645	21.104
	CycleGAN [24]	358.90	0.58697	17.260	20.706
	SPADE [34]	435.26	0.76502	19.297	20.734
	SelectionGAN [35]	406.55	0.62731	16.811	21.092
	TSIT [36]	349.14	0.56122	15.940	20.555
	LPTN [37]	340.72	0.64413	17.951	20.653
	SMAPGAN [3]	464.65	0.72831	18.655	**21.624**
	CreativeGAN [5]	309.06	0.44415	14.517	21.547
	Ours	**299.29**	**0.42323**	**14.177**	21.584
16	Pix2Pix [23]	308.65	0.49493	15.590	23.490
	Pix2PixHD [2]	251.64	0.40092	14.694	23.118
	CycleGAN [24]	263.18	0.44191	15.172	22.646
	SPADE [34]	341.83	0.63769	16.710	23.167
	SelectionGAN [35]	330.58	0.50611	15.194	23.373
	TSIT [36]	278.61	0.46925	14.655	22.670
	LPTN [37]	307.11	0.55695	16.345	22.824
	SMAPGAN [3]	381.78	0.64536	16.991	**24.254**
	CreativeGAN [5]	205.22	0.30197	12.805	23.609
	Ours	**191.48**	**0.29212**	**12.248**	23.545
17	Pix2Pix [23]	269.05	0.40735	13.849	24.426
	Pix2PixHD [2]	206.56	0.38304	14.263	23.434
	CycleGAN [24]	200.38	0.40405	14.214	23.228
	SPADE [34]	326.26	0.60956	15.703	24.148
	SelectionGAN [35]	290.40	0.46775	14.826	24.216
	TSIT [36]	227.45	0.46671	14.555	23.381
	LPTN [37]	235.61	0.44371	14.484	23.015
	SMAPGAN [3]	352.25	0.59993	16.116	**25.133**
	CreativeGAN [5]	171.24	**0.32882**	**13.253**	23.815
	Ours	**165.85**	0.33886	13.333	24.124
18	Pix2Pix [23]	162.52	**0.29129**	**11.718**	25.158
	Pix2PixHD [2]	162.87	0.39792	13.804	23.981
	CycleGAN [24]	146.50	0.39239	13.335	23.999
	SPADE [34]	254.02	0.55138	14.127	25.184
	SelectionGAN [35]	252.53	0.49446	14.607	24.831
	TSIT [36]	189.61	0.49031	14.364	24.161
	LPTN [37]	192.57	0.46029	14.539	22.571
	SMAPGAN [3]	307.02	0.61090	15.059	**26.285**
	CreativeGAN [5]	138.27	0.38752	13.495	24.210
	Ours	**142.77**	0.39634	13.658	24.807
Avg.	Pix2Pix [23]	284.06	0.44718	14.689	23.548
	Pix2PixHD [2]	253.68	0.42968	14.851	22.909
	CycleGAN [24]	242.24	0.45633	14.995	22.645
	SPADE [34]	339.34	0.64091	16.459	23.308
	SelectionGAN [35]	320.01	0.52391	15.360	23.378
	TSIT [36]	261.20	0.49687	14.878	22.692
	LPTN [37]	269.01	0.52627	15.830	22.266
	SMAPGAN [3]	376.42	0.64612	16.705	**24.324**
	CreativeGAN [5]	205.95	0.36561	13.518	23.295
	Ours	**199.85**	**0.36264**	**13.354**	23.515

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TABLE XV
QUANTITATIVE SINGLE-LEVEL RESULTS ON AIDOMG. THE RESULTS OF OUR METHOD HAVE HIGHER QUALITY THAN OTHER METHODS

City	Method	FID↓	KMMD↓	WD↓	PSNR↑
Haikou	Pix2pix [23]	148.03	0.38474	12.067	27.627
	Pix2pixHD [2]	168.94	0.39606	12.006	27.361
	CycleGAN [24]	112.14	0.27461	10.428	26.484
	SPADE [34]	308.19	0.70027	15.144	27.135
	SelectionGAN [35]	248.57	0.53087	13.383	27.645
	TSIT [36]	151.65	0.40315	11.554	26.641
	LPTN [37]	213.91	0.51806	13.402	26.729
	SMAPGAN [3]	324.78	0.73765	15.644	**29.839**
	CreativeGAN [5]	133.13	0.29809	10.653	27.588
	Ours [2]	**105.07**	**0.23826**	**9.810**	27.657
Paris	Pix2pix [23]	96.12	0.2062	9.450	26.462
	Pix2pixHD [2]	125.19	0.2405	9.811	26.814
	CycleGAN [24]	132.31	0.2340	9.218	25.508
	SPADE [34]	276.79	0.5367	12.863	26.744
	SelectionGAN [35]	254.71	0.4833	12.135	27.254
	TSIT [36]	149.74	0.3425	10.287	25.872
	LPTN [37]	216.35	0.4672	12.264	25.778
	SMAPGAN [3]	287.60	0.6138	13.811	**28.984**
	CreativeGAN [5]	81.93	0.1441	8.577	26.961
	Ours [2]	**72.90**	**0.1403**	**8.488**	27.078

geographic elements of our various levels are more complete, and the metrics are greatly improved.

3) MLF Generator w/o the $(k-1)$*th Level:* Without the $(k-1)$th level inputs, the model is equivalent to a lightweight generator. It uses the preliminary generated map to further get the refined map. It is one of the components of the dual-generator architecture and is used to tackle the large domain gap. As shown in Fig. 7(e), we can see that after adding the MLF generator, the visual effect as a whole is improved. The details of the geographic elements are complemented. From Table XI, it can be found that all the metrics are improved after adding the generator, especially FID is increased by 24.36. By adding the MLF generator w/o the $(k-1)$th level, the preliminary map is reprocessed to make it closer to the real map.

4) MLF Generator w/ the $(k-1)$*th Level:* The purpose of adding the MLF generator w/ the $(k-1)$th level is to fuse the map information of the lower levels. It is used to further ensure the main content consistency. As shown in Fig. 7(f), after considering multilevel information, we can reduce some misjudgments of geographic elements caused by considering only this level of data. Multilevel information can improve the quality of the generated map. From Table XI, we can see that after adding the MLF generator w/ the $(k-1)$th level, all the results are improved compared without adding the $(k-1)$th level information, especially PSNR is increased by 0.129. By adding the MLF generator w/ the $(k-1)$th level, the preliminary map incorporates information from other levels and is processed to obtain a higher quality refined map.

E. Comparison of the Different Format of Level Label

We conduct experiments to find the appropriate level label format. As shown in Table XII, we can find that for level labels, the one-hot format level label is more conducive to model understanding and achieves the best results. Compared with the one-hot format level label, the numerical format level label is more difficult to be understood by the model, so the results are worse. The level classifier is expected to clarify the level to which the map belongs, rather than have the generalization ability to classify maps. Therefore, the result of using soft label is not as good as that of one-hot format level label.

F. Comparison of Whether Auxiliary Function Is Used

We conduct experiments to compare the gain in results using auxiliary functions. As shown in Table XIII, we can find that the results with auxiliary functions are better than those without auxiliary functions in most metrics, which proves the correctness of using auxiliary functions in ME extractor.

G. Generalization Analysis of Cross Study Areas

To verify the generalization ability of our method in different regions, we use the SI images from the CN region for training and the SI images from the US region for testing. Table XIV shows the evaluation results of the relevant methods. From Table XIV, we can see that our method achieves the best results on the GAN metrics averaged across levels. It means that our method is still optimal for the SI-based MLMG across regions.

H. Comparison of the SI-Based Single-Level Map Generation

The SI-based single-level map generation is a special case of the SI-based MLMG. To verify the generation effect of our method on the single-level map, we conduct single-level generation experiments on AIDOMG [5] which is the latest single-level map public dataset. We follow [5] to experiment on the cities Haikou and Paris, where Haikou is relatively complex and Paris is relatively simple. Table XV shows the quantitative results of the related methods on AIDOMG, and we can see that our method also maintains the optimal results for the SI-based single-level map generation. These further demonstrate the superiority of our method.

VI. CONCLUSION

In this article, we analyze the challenges of the SI-based MLMG, including the large domain gap, level-dependent content differences, and main content consistency. We propose a level-aware fusion network for the SI-based MLMG, which aims to solve the challenges of the SI-based MLMG. To tackle the large domain gap, we propose to generate maps in a coarse-to-fine way, i.e., the RD generator and MLF generator form a dual-generator structure. We use the level classifier to discriminate the level of the map, which ensures the level-dependent content differences. We use the ME extractor to provide the major element features and the MLF generator to fuse the multilevel preliminary map, which are used to ensure the main content consistency. Our method achieves the best performance compared with the state-of-the-art SI-based map generation methods and image-to-image translation methods. In addition, to satisfy the needs of multilevel map training and testing, we collect a level-balanced large dataset MLMG from Google Maps. It provides 18 700 samples

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of SI images, corresponding map images, level labels, and element labels.

In the existing results, some geographical items still lose their topology information, e.g., the road is not very consistent. In the future, we plan to investigate effective geometric constraints to further improve our model.

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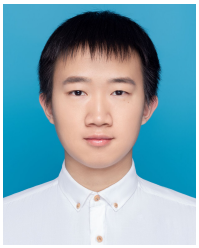

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